INFORMATION SYSTEMS 3A
INSY7311

Learning Unit 7: Evaluation and Usability Testing

With

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Lecture 10

IIE Bachelor of Computer and Information Sciences in Application Development (BCA)

Leaning Unit Objectives

Learning Unit 7: Evaluation and Usability Testing

Theme 1: Evaluation

LO1: Describe what evaluation means in HCI/Usability.

LO2: Explain when to use quantitative and qualitative approaches in design.

Theme 2: Expert Reviews

LO3: Define expert reviews and why they are necessary.

LO4: Explain the different types of expert reviews.

Theme 3: Usability testing

LO5: Discuss the importance of usability testing in the design process.

LO6: Explain the different types of usability tests.

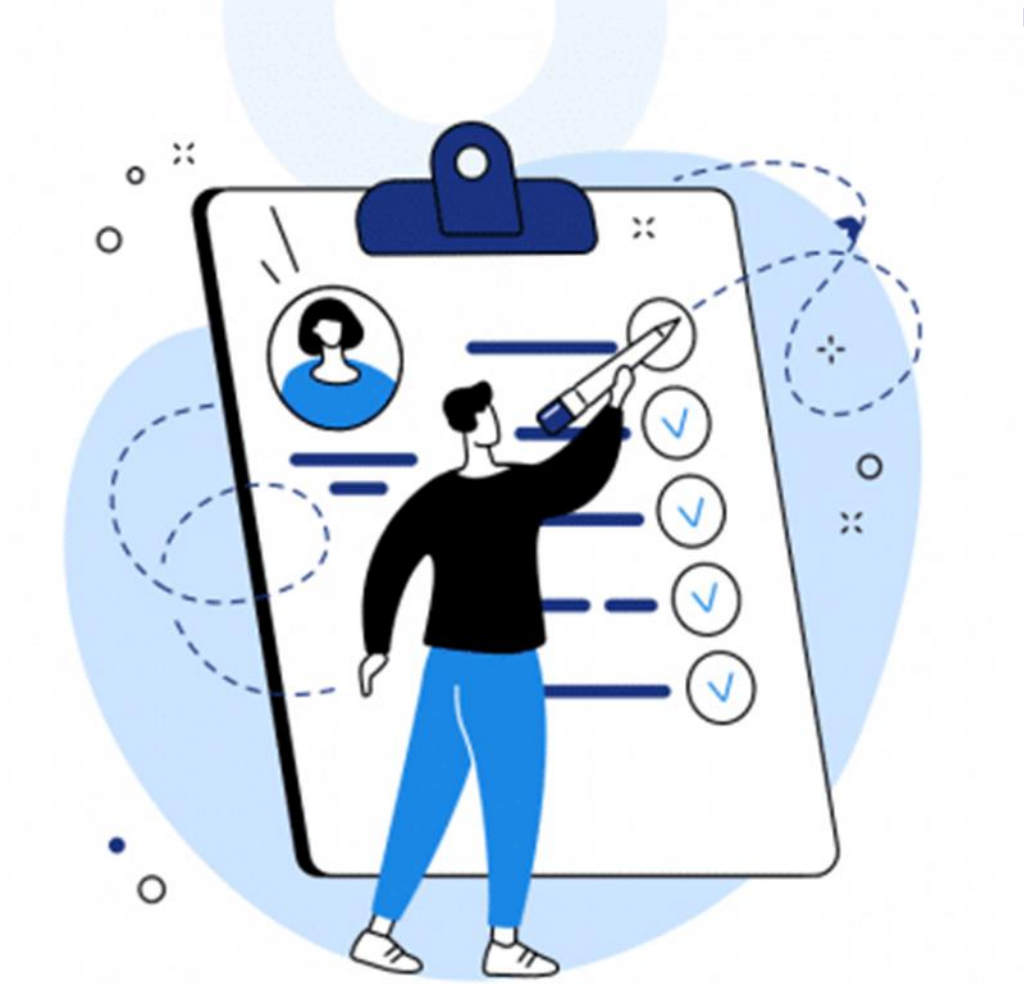
Theme 1: Evaluation

Evaluation

What Evaluation Means in HCI/Usability

Definition of Evaluation in HCI

- Evaluation in Human-Computer Interaction (HCI) refers to:
- *The systematic process of **collecting and analyzing data about users' interactions with a system or prototype** in order to improve its design.*



Evaluation

Key Purpose of Evaluation

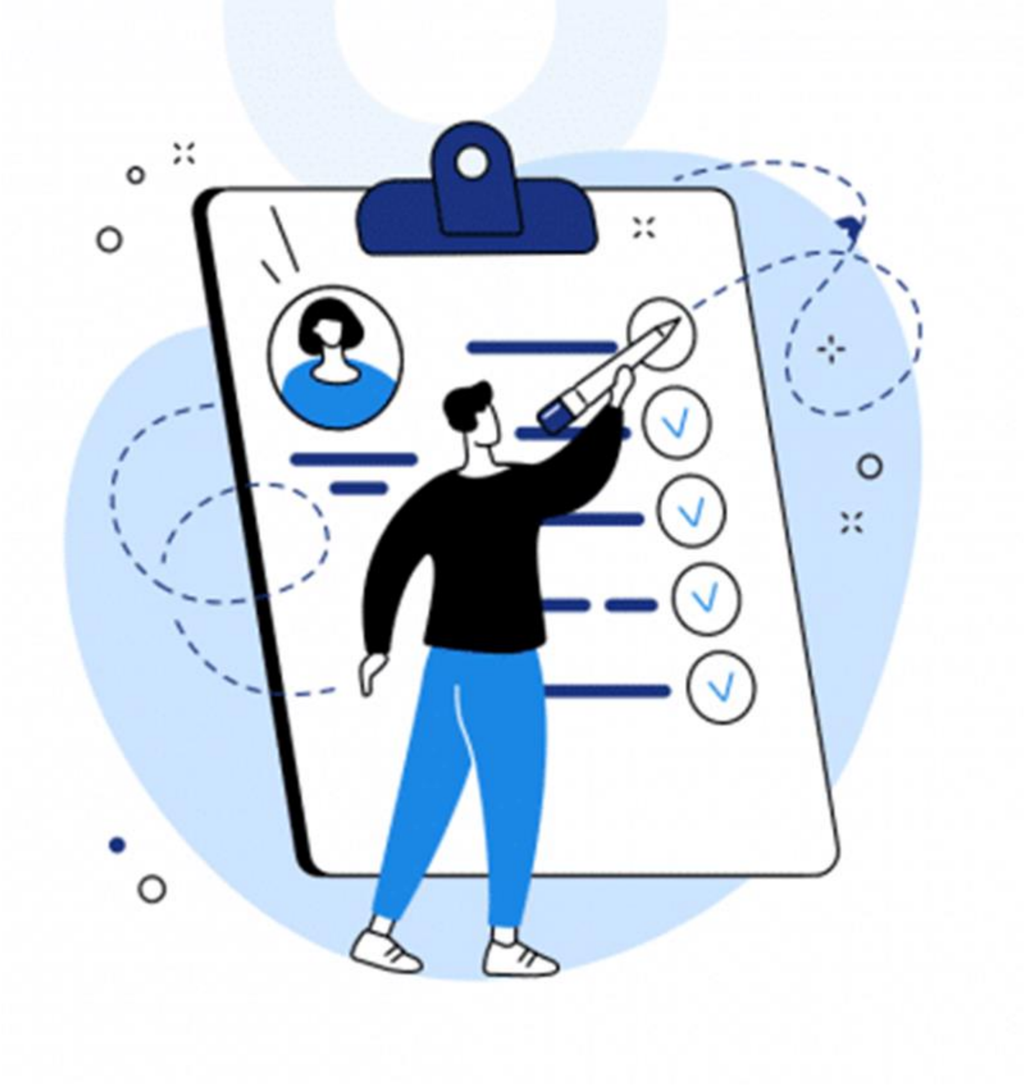
- Evaluation is not a one-time activity, it is **continuous throughout the design lifecycle.**

Its primary goals are to:

- Assess **usability** (ease of use, efficiency, learnability)
- Understand **user experience (UX)** (satisfaction, enjoyment, engagement)
- Identify **design problems**
- Provide **evidence-based improvements**

Evaluation focuses on both:

- **Usability** → How easy the system is to use
- **User experience** → How users feel when interacting with it



Evaluation

Why Evaluation is Important

Evaluation prevents a critical mistake in design:

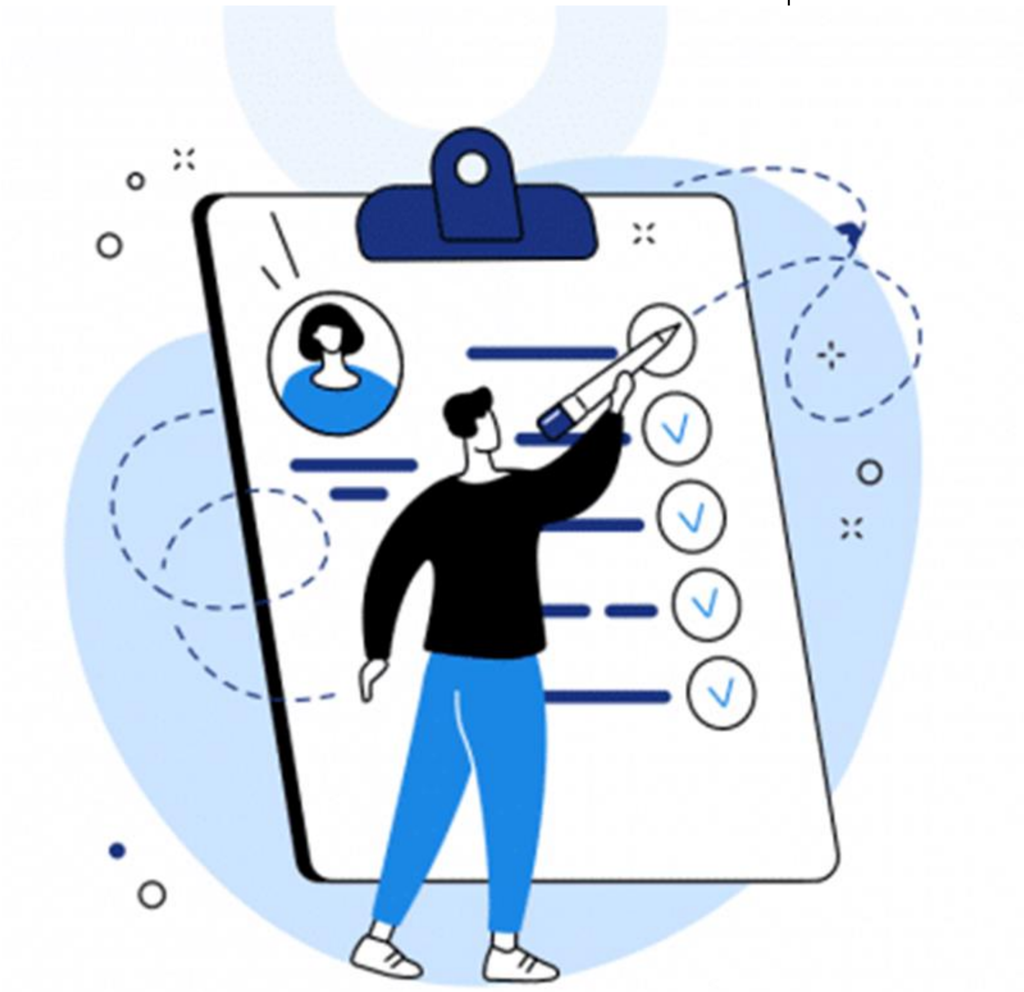
- Designing for yourself instead of the actual users.

Without evaluation:

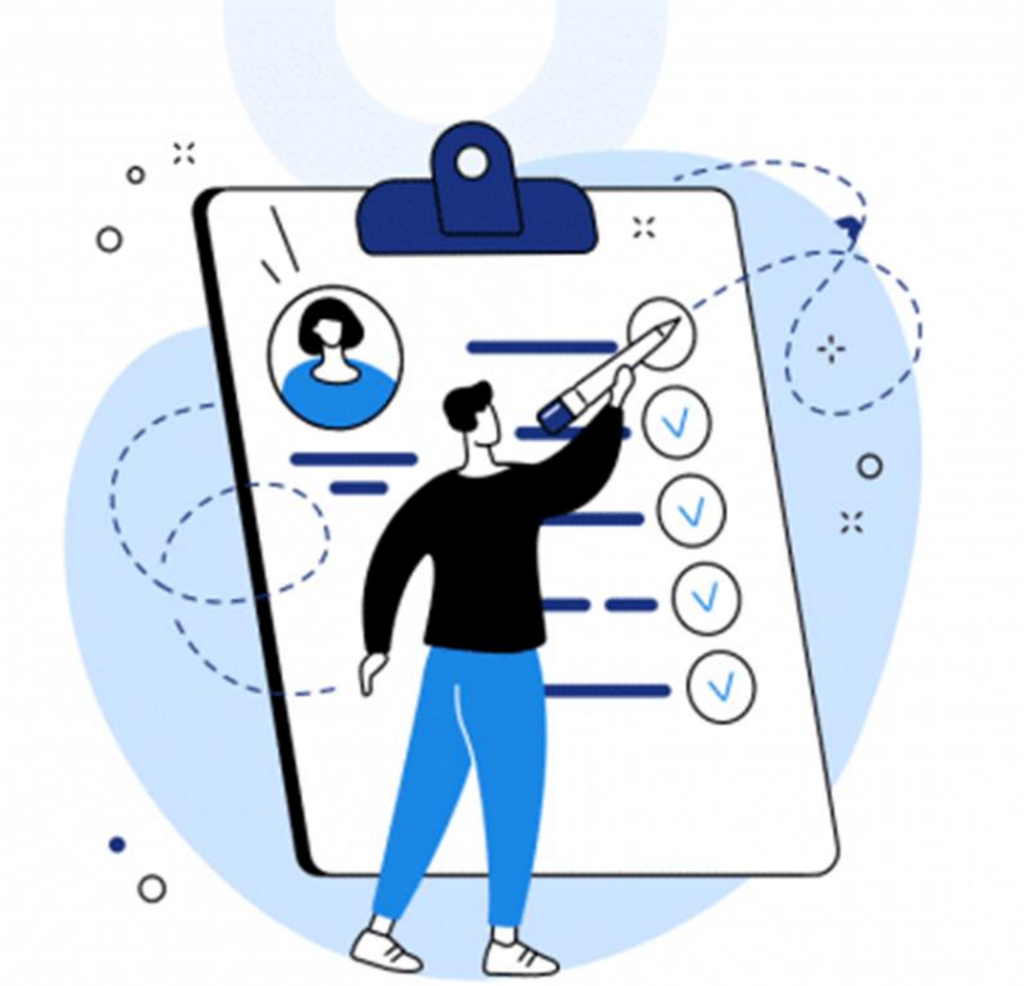
- Designers rely on assumptions
- Systems may fail real-world use
- Usability problems remain hidden

With evaluation:

- Designs are **validated**
- Decisions are **data-driven**
- Systems become **user-centered**



Evaluation



Where and When Evaluation Happens

Evaluation can occur:

- **Early stage** → Concepts and sketches
- **Mid stage** → Prototypes
- **Late stage** → Fully developed systems

It can also take place in different environments:

- **Controlled environments (labs)**
- **Real-world settings (field studies)**
- **Remote/online environments**

When to Use Quantitative and Qualitative Approaches

Evaluation relies on two major types of data:

Quantitative Approaches

Definition

- Numerical data used to measure performance.

Examples (from usability testing):
- Time taken to complete a task
- Number of errors made
- Task success rate
- Number of clicks or interactions

When to Use Quantitative Methods

Use quantitative methods when:

- You need **objective measurement**
- Comparing **different designs**
- Evaluating **efficiency and performance**
- Testing **hypotheses**

Advantages

- Easy to analyze statistically
- Enables comparison across users
- Provides measurable benchmarks

When to Use Quantitative and Qualitative Approaches

Qualitative Approaches

Definition

- Non-numerical data that explains **why users behave the way they do.**

Examples

- User interviews
- Observations
- Think-aloud protocols
- User feedback and opinions

When to Use Qualitative Methods

Use qualitative methods when:

- Exploring **user feelings and experiences**
- Understanding **user behavior**
- Identifying **hidden usability issues**
- Designing **new systems**

When to Use Quantitative and Qualitative Approaches

Combining Both Approaches

- Effective evaluation combines both quantitative and qualitative data for deeper insight.

Example

- Quantitative: Users take 2 minutes to complete a task
- Qualitative: Users feel confused during navigation

Together, they provide a **complete understanding**

Theme 2:
Expert Reviews

Expert Reviews and Why They Are Necessary

What Are Expert Reviews?

Expert reviews (also called **inspection methods**) involve:

- *Specialists evaluating an interface **without involving real users,** using their knowledge of usability principles.*

Expert Reviews and Why They Are Necessary

Why Expert Reviews Are Necessary

Early Detection of Problems

- Identify issues before user testing
- Saves time and cost

No Need for Users

Useful when users are:

- Unavailable
- Expensive to recruit
- Hard to access

Fast and Cost-Effective

- Quick feedback on prototypes
- Ideal for early-stage designs

Complement Usability Testing

- Experts find different types of problems
- Used alongside user testing for better results

Expert Reviews and Why They Are Necessary

When to Use Expert Reviews

- Early design stages
- During rapid prototyping
- Before usability testing
- When resources are limited

Types of Expert Reviews

Heuristic Evaluation

Definition

- Experts evaluate a system based on **usability principles (heuristics).**

Key Idea

- Check whether the interface follows established usability rules.

Examples of Nielsen's Heuristics:

- Visibility of system status
- Consistency and standards
- Error prevention
- User control and freedom

Strengths

- Fast and structured
- Widely used in industry
- Identifies common usability issues

Types of Expert Reviews

Cognitive Walkthrough

Definition
- Experts simulate a user’s problem-solving process step-by-step.

Focus
- How easy it is for **new users to learn the system**

Key Questions
- Will the user know what to do?
- Will they notice the correct action?
- Will they understand the feedback?

Strengths
- Good for learnability
- Task-focused evaluation

Types of Expert Reviews

Pluralistic Walkthrough (Extension)

Involves **multiple stakeholders**

- Designers
- Developers
- Users

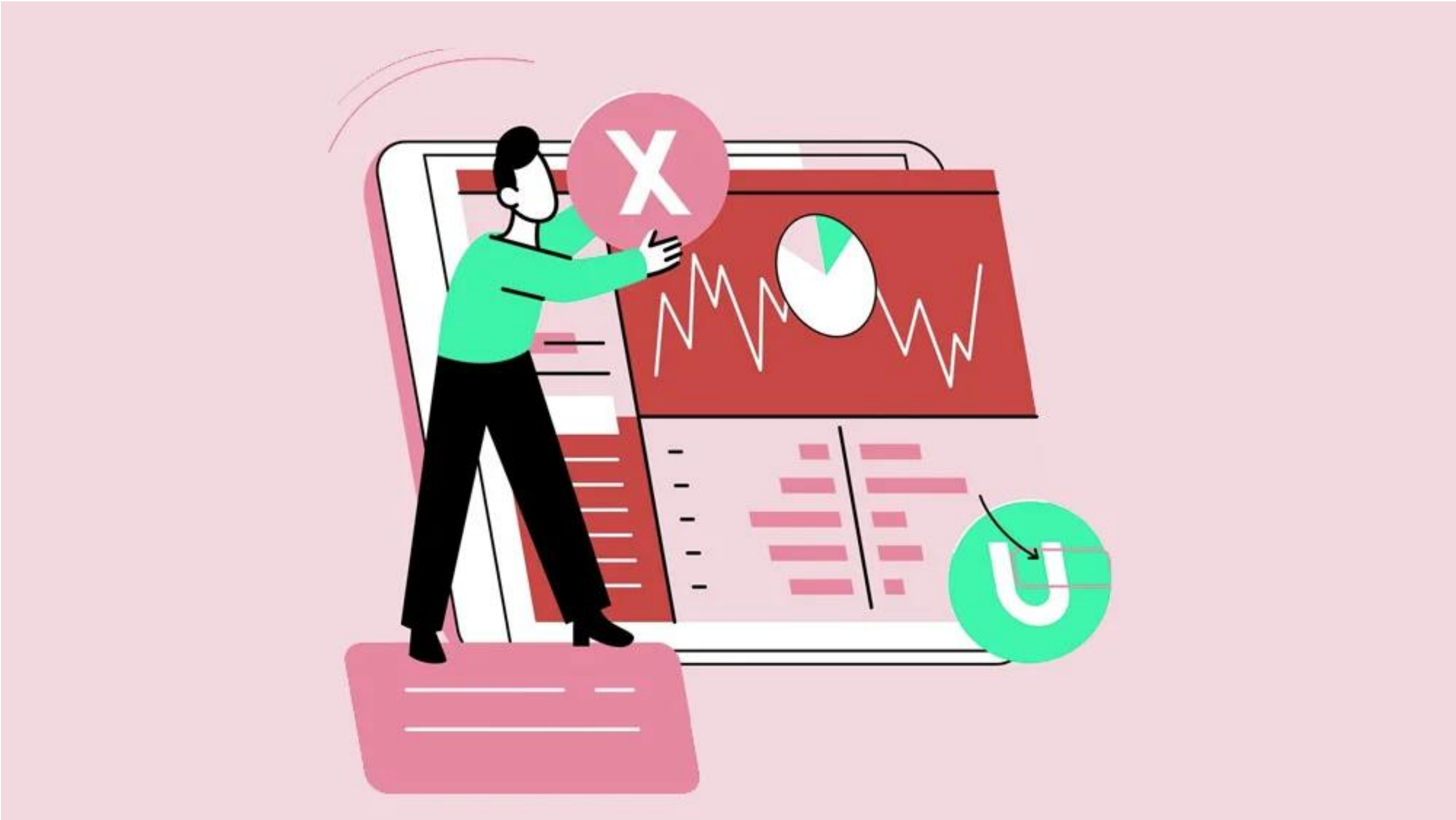
Encourages **discussion and multiple perspectives**

Types of Expert Reviews

Summary of Expert Reviews

Method	Focus	Strength
• Heuristic Evaluation	General usability	Fast, structured
• Cognitive Walkthrough	Learnability	Task-focused
• Pluralistic Walkthrough	Collaboration	Multiple perspectives

Theme 3:
Usability Testing



Importance of Usability Testing in the Design Process

Definition of Usability Testing

Usability testing is:

- The process of observing real users performing tasks to evaluate how usable a system is.

Core Objective

To determine whether:

- Users can **complete tasks successfully**
- The system is **efficient**
- The experience is **satisfying**

Importance of Usability Testing in the Design Process

Usability Testing is Critical

Real User Validation
- Tests with **actual users**
- Reveals real-world problems

Measures Performance
- Task completion time
- Error rates
- Success rates

Improves User Experience
- Identifies frustration points
- Enhances satisfaction

Supports Iterative Design
- Test → Improve → Test again

Importance of Usability Testing in the Design Process

What Happens During Usability Testing

Users are asked to:

- Perform predefined tasks
- Interact with the system

Data collected includes:

- Task performance (time, errors)
- User behavior (clicks, navigation)
- User thoughts (think-aloud)
- Feedback (interviews, questionnaires)

Importance of Usability Testing in the Design Process

Types of Data Collected

Quantitative

- Time to complete task
- Number of errors
- Success rate

Qualitative

- User opinions
- Observations
- Emotional reactions

Types of Usability Tests

Usability testing can be classified based on **environment and approach:**

Laboratory Testing (Controlled Environment)

Description

- Conducted in usability labs
- Controlled conditions

Advantages

- High control over variables
- Accurate measurement

Disadvantages

- Artificial environment
- May not reflect real-world use

Types of Usability Tests

Field Testing (Natural Environment)

Description

- Conducted in real-world settings (homes, offices)

Advantages

- Realistic context
- Captures natural behavior

Disadvantages

- Less control
- Harder to manage

Types of Usability Tests

Remote Usability Testing

Description

- Conducted online (Zoom, Teams, etc.)

Advantages

- Cost-effective
- Access to wider participants

Disadvantages

- Less control over environment

Types of Usability Tests

Think-Aloud Testing

Description

- Users verbalize their thoughts while performing tasks.

Purpose

- Understand user reasoning
- Reveal hidden issues

Types of Usability Tests

A/B Testing (Experimental Testing)

Description

- Compare two versions of a design
- Users randomly assigned

Purpose

- Determine which design performs better

Types of Usability Tests

Summary of Usability Testing Types

Type	Environment	Purpose
• Lab Testing	Controlled	Performance measurement
• Field Testing	Real-world	Contextual understanding
• Remote Testing	Online	Accessibility
• Think-Aloud	Any	User thinking
• A/B Testing	Controlled/Online	Compare designs

Conclusion

- Evaluation and usability testing are **core pillars of user-centered design.**

Key Takeaways

- Evaluation is **continuous and iterative**
- Both **quantitative and qualitative data** are essential
- Expert reviews provide **early insights**
- Usability testing validates designs with **real users**
- Combining methods leads to **better design decisions**

Next Class:

Learning Unit 8: Interaction styles and Interactive Visualisation

Thank you for listening

Any question?